DSML Module Test: ML 1.2 : Adv Supervised Algorithms (No Class Today)

Performance Report

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DSML Module Test: ML 1.2 : Adv Supervised Algorithms (No Class Today)

100.0
/100

SCORE

Duration : 2 hours

Total Questions : 11

Attempted : 11

Solved : 11

Start Time : Sunday, 16 Aug 2026 2:44 PM IST

QUESTION STATS

S1 Default Section (1050/1050)

#
Topic ⓘ
Type/Language ⓘ

Score ⓘ
Status ⓘ
Actions
No:
1
Problem:
K-Means
Type/Language:
</>
Python 3 (python-3.8)
Score:
100.0 / 100.0
Status:
Solved
View Details

K-Means using manhattan distance

Given the data, perform k-means clustering using **manhattan distance** and print the updated centroids for each of the 5 iterations of k-means. The centroids are a NumPy array rounded off to two decimal places.

The manhattan distance between two points of dimensions k, $d(p, q)$ = $\sum *{i=1}^{k} \left| q*{i}-p_{i}\right| $

Input Format:

```
Two lists are taken as input where the first line has points and the second line has a list of initi
alized centroids which are then typecasted into numpy 2d arrays.
```

Output Format:

```
The 2d NumPy array consisting of centroids are printed for 5 iterations
```

Sample Input:

points = [(10,10), (11,11), (12,12), (15,15),(1,2),(3,4),(1,4),(0,4),(0,5),(-2,4),(-3,5),(6,-8),(5,6),(-1,-2),(3,-5),(4,-8),(5,-10),(-4,-4),(-2,-1),(0,0),(-3,0),(3,3),(4,4),(1,2),(6,7),(-9,0)])
initial centroids = [(1,1),(-1,1),(-1,-1)]

Sample Output:

```
[[ 2.8   0.67]
 [-4.25  2.25]
 [-2.33 -2.33]]
[[ 3.23  0.08]
 [-2.8   3.6 ]
 [-2.5  -1.75]]
[[ 3.42 -0.25]
 [-0.8   4.4 ]
 [-3.8  -1.4 ]]
[[ 3.57 -3.43]
 [ 1.2   4.3 ]
 [-3.8  -1.4 ]]
[[ 4.5  -7.75]
 [ 1.58  4.17]
 [-3.17 -1.17]]
```

User Code

```
 import numpy as np

points = np.array(eval(input()))
centroids = np.array(eval(input()))

#function returns the manhattan distance between points p1 and p2
def calc_dist(p1, p2):
    """p1 and p2 are two tuples representing the points"""

    dist = 0

    #YOUR CODE GOES HERE
    dist = np.sum(np.abs(np.array(p1) - np.array(p2)))

    return dist


#function returns the updated centroid for the points in lst which represent a cluster
def update_centroid(lst):
    """lst is a 2d numpy array consisting of points in a cluster"""

    new_centroid = None

    #YOUR CODE GOES HERE
    new_centroid = np.mean(lst,axis=0)

    return new_centroid

#function performs one iteration of k-means
def k_means(points, centroids):
    """points is a 2d numpy array consisting of points whereas
    centroids is a 2d numpy array consisting of initial centroids"""


    # clusters[i] stores all the points with the cluster label i
    clusters = [[] for i in range(len(centroids))]
```

```
    for pnt in points:
        #tmp[i] stores the distance of pnt from centroid[i]
        tmp = [0]*len(centroids)
        for i,centroid in enumerate(centroids):
            tmp[i] = calc_dist(pnt,centroid)   #calculate the distance between point and centroid

        cluster_id = np.argmin(tmp)  #index of the centroid from which pnt is closest
        clusters[cluster_id].append(pnt)


    #update the centroid of each cluster
    new_centroid = []
    for cluster in clusters:
        new_centroid.append(update_centroid(cluster)) #update the centroid of cluster using update_centroid f
unction we defined earlier

    return np.array(new_centroid)

for i in range(5):
    centroids = k_means(points,centroids)  #perform ith iteration on data to update clusters using k_means fu
nction we defined earlier
    print(np.round(centroids,2))
```

Previous Submissions:

#	Input Type	Time of submission	Result	Time	Memory	Score	Source	Submission Mode

1	Correctness	Sun, 16 Aug 2026 14:58:30 +0530	**Correct Answer**	523 ms	59068 KB	**50.0**	**View Code**	**Live Mode**

No:
2
Problem:
clustering
Type/Language:

Score:
100.0 / 100.0
Status:
Solved
View Details

Value of Dunn index

A higher Dunn Index value indicates:

- (●) **Better cluster separation and compactness.**
- ( ) **Higher noise levels in the data.**

- ○ **Poor clustering quality.**
- ○ **A need for more clusters.**

No:
3
Problem:
clustering
Type/Language:
Score:
100.0 / 100.0
Status:
Solved
View Details

Elbow point

In the Elbow Method graph, what does the “elbow point” represent?

- ○ **The point where the algorithm converges.**

○ **The point where the sum of squared distances becomes zero.**

○ **The point where the silhouette score reaches its maximum.**

◉ **The optimal number of clusters where the rate of decrease in sum of squared distances slows down.**

No:
4
Problem:
clustering
Type/Language:
Score:
100.0 / 100.0
Status:
Solved
View Details

Merging strategy in agglomerative

In agglomerative hierarchical clustering, which merging strategy starts by treating each data point as an individual cluster and then repeatedly merging the closest clusters until the desired number of clusters is achieved?

- ○ **Complete linkage**
- ○ **Average linkage**
- ◉ **Single linkage**
- ○ **Ward linkage**

No:
5
Problem:
DSML Advanced : Anomaly & Outlier Detection
Type/Language:

Score:
100.0 / 100.0
Status:
Solved

View Details

Assumption of Elliptic Envelope

The Elliptic Envelope method assumes that the inlying data follows:

- ○ **Linear distribution.**
- ◉ **Normal distribution.**
- ○ **Uniform distribution.**
- ○ **Exponential distribution.**

No:
6
Problem:
DSML Advanced : Anomaly & Outlier Detection
Type/Language:
Score:
100.0 / 100.0
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Solved

View Details

Value of LOF

In the context of the Local Outlier Factor (LOF) method, a data point with an LOF value greater than 1 indicates that the point:

- ○ **Is an inlier within a dense cluster.**
- ◉ **Is an outlier with low local density.**
- ○ **Deviates significantly from the mean.**
- ○ **Has few neighbors within the epsilon radius.**

No:
7
Problem:
t-SNE
Type/Language:
Score:
100.0 / 100.0
Status:

Solved

View Details

t-SNE in high dim

Why is t-SNE particularly effective in visualizing clusters of high-dimensional data?

- ○ **It creates a linear mapping of the data.**
- ○ **It amplifies the variance of the data.**
- ○ **It focuses on the global structure of the data.**
- ◉ **It preserves local pairwise relationships.**

No:
8
Problem:
Miscellaneous
Type/Language:

Score:
50.0 / 50.0
Status:
Solved

View Details

Ideal Clustering

For an ideal cluster, which of the following statements is true?

- Inter cluster distance should be high
- Inter cluster distance should be low
- Intra cluster distance should be high
- Intra cluster distance should be low

- ○ **1,3**
- ○ **2,4**
- ◉ **1,4**
- ○ **3,4**

No:

9

Problem:

Miscellaneous

Type/Language:

Score:

100.0 / 100.0

Status:

Solved

View Details

K-means initialisation

Which of the following can arise from poor initialization in K-Means clustering?

- ○ **Slower convergence to the optimal solution**
- ○ **Sensitivity to the number of clusters (K)**
- ◉ **Increased risk of convergence to a local minimum**
- ○ **Improved robustness against outliers**

No:
10
Problem:
Miscellaneous
Type/Language:

Score:
100.0 / 100.0
Status:
Solved
View Details

DBSCAN Noisy Pts

How does DBSCAN handle noisy data points?

- ○ **By assigning noisy points to the nearest cluster**
- ◉ **By identifying core points, border points, and treating isolated points as noise**

○ **By performing outlier removal before clustering**

○ **By creating a separate cluster for noisy points**

No:
11
Problem:
Miscellaneous
Type/Language:

Score:
100.0 / 100.0
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Solved

View Details

PCA Steps

Which of the following are steps involved in performing PCA?

- Scaling the features to a similar range.
- Compute the cosine similarity between each pair of data points.
- Calculating the covariance matrix of the data.

- Choosing the optimal number of clusters.

- ○ 1,2
- ○ 1,4
- ○ 2,3
- ◉ 1,3

 **Want to attempt again in Practice mode?** Retry Academy contest (https://scaler.com/test/77f53ef3c5/?signature=BAhpA7iWcw%3D%3D--9f26aceda4260a775c73405865a6302e2efedc05)